

CAMAF: Context-Aware Multimodal Alignment Framework for explainable lung disease risk stratification

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ARTICLE INFO

Keywords:

Lung disease
Multimodal machine learning
Risk stratification
Explainability
Dynamic alignment

ABSTRACT

Lung disease remains a leading cause of morbidity and mortality worldwide, necessitating early and accurate risk stratification to improve patient outcomes. Existing approaches rely on either clinical data or imaging modalities independently, limiting their ability to capture the intricate interactions between structured and unstructured information. To overcome this limitation, we propose the Context-Aware Multimodal Alignment Framework (CAMAF), a novel AI-driven approach that dynamically integrates structured clinical data from the MIMIC-IV dataset and unstructured imaging data from the CheXpert dataset for context-aware and interpretable lung disease risk stratification. Unlike conventional multimodal fusion techniques, CAMAF introduces a Context-Gated Attention (CGA) mechanism, which adaptively aligns modality contributions based on patient-specific contexts. Additionally, CAMAF employs TabTransformer and Vision Transformer (ViT) for high-fidelity feature extraction and leverages a Transformer Encoder for enhanced multimodal fusion, ensuring superior predictive performance. Experimental results demonstrate the efficacy of CAMAF, achieving 92.78% accuracy and 0.9457 AUC-ROC, outperforming state-of-the-art machine learning baselines and traditional clinical risk scores (e.g., CURB-65, BODE Index). Robustness analysis highlights its ability to maintain performance under noisy and missing data conditions, while SHAP and Grad-CAM provide interpretable explanations for clinical decision-making. Despite its advantages, CAMAF faces challenges in handling rare pathologies, which can be addressed through data augmentation and transfer learning. This work establishes CAMAF as a novel, clinically viable AI framework, bridging the gap between multimodal data integration, interpretability, and real-world clinical applicability in lung disease risk stratification.

1. Introduction

Lung diseases remain a major global health burden, contributing significantly to mortality and morbidity worldwide (Olsson et al., 2023). Conditions such as chronic obstructive pulmonary disease (COPD), pneumonia, and lung cancer rank among the leading causes of death, with millions of cases reported annually (Al-qaness et al., 2024; Murguia & Gambelunghe, 2022). These diseases not only impose substantial healthcare costs but also demand effective strategies for early diagnosis and risk stratification to prevent adverse outcomes. Accurate risk stratification can enable timely clinical interventions, reduce hospitalizations, and optimize resource allocation, particularly in resource-constrained settings (Soffer et al., 2022).

Recent advancements in machine learning have demonstrated substantial potential in healthcare applications, including diagnostic assistance (Goyal & Singh, 2023), prognosis prediction (Dack et al., 2023), and personalized treatment planning (Cravo et al., 2022). However, many existing models rely on unimodal data, such as structured

electronic health records (EHRs) (Ros & Viejo, 2024) or unstructured imaging data like chest X-rays (Rao, Rajitha, Srinivasu, Ijaz, & Woźniak, 2024). This unimodal focus limits the models' ability to capture the complex, multifactorial nature of lung diseases, which often involves interactions between physiological, temporal, and spatial factors. Moreover, the lack of explainability in traditional machine learning frameworks hinders their clinical adoption, as healthcare providers require transparent and interpretable models to build trust and validate predictions.

The integration of structured clinical data and unstructured imaging data offers a promising avenue for improving predictive accuracy and enhancing the interpretability of machine learning models (Cravo et al., 2022; Goyal & Singh, 2023; Ros & Viejo, 2024). Clinical data provides vital information on patient demographics, comorbidities, and temporal trends, while imaging data captures spatial patterns and abnormalities indicative of lung pathology (Viswanathan, Toro, Corredor, Mukhopadhyay, & Madabhushi, 2022). Despite these advantages,

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<https://doi.org/10.1016/j.eswa.2025.127398>

Received 1 February 2025; Received in revised form 18 March 2025; Accepted 21 March 2025

Available online 31 March 2025

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the development of multimodal frameworks is challenging due to the heterogeneity of modalities, the dynamic nature of clinical data, and the need for interpretable outputs. While multimodal approaches have shown potential, many existing frameworks fail to adaptively weigh the contributions of each modality based on patient-specific contexts. Static fusion methods often overlook temporal variations in clinical features and the localized nature of imaging abnormalities, limiting their effectiveness. Additionally, the lack of interpretability in many models poses a significant barrier to clinical adoption. These limitations highlight the need for a robust, adaptive, and interpretable framework capable of leveraging the complementary strengths of structured and unstructured data for accurate lung disease risk stratification. This research introduces the Context-Aware Multimodal Alignment Framework (CAMAF), which addresses the aforementioned challenges. The key contributions of this work are as follows:

- Developed a novel multimodal framework that integrates structured clinical data (MIMIC-IV) and unstructured imaging data (CheXpert) using TabTransformer and Vision Transformer (ViT), respectively.
- Proposed a Context Encoder and Context-Gated Attention mechanism to dynamically align clinical and imaging embeddings based on patient-specific contexts, ensuring relevance and robustness.
- Achieved superior performance, with an accuracy of 92.78% and an AUC-ROC of 0.9457, significantly outperforming state-of-the-art baselines in lung disease risk stratification.
- Enhanced model explainability through SHAP for clinical data and Grad-CAM for imaging data, providing actionable insights into key features and imaging regions influencing predictions.
- Demonstrated robustness to noise and incomplete data, validating the framework's reliability in real-world scenarios.

The proposed CAMAF framework not only advances the state of the art in multimodal machine learning but also provides a clinically interpretable solution for lung disease risk stratification, paving the way for improved patient outcomes and informed decision-making. The paper is organized as follows. Section 2 reviews related works, discussing existing methodologies for lung disease risk stratification and their limitations. Section 3 presents the proposed Context-Aware Multimodal Alignment Framework (CAMAF), detailing its components, including feature embedding, context encoding, dynamic alignment, fusion mechanisms, and the classification layer. Section 4 evaluates the framework's performance through experimental comparisons with baseline models, sensitivity analyses, ablation studies, robustness tests, and explainability assessments using SHAP and Grad-CAM. Section 5 outlines the limitations of the current approach, such as challenges in data heterogeneity, model complexity, and generalizability, providing directions for future research. Section 6 concludes the paper by summarizing the contributions, results, and potential applications of CAMAF in healthcare.

2. Related works

This section provides an overview of existing research in lung disease risk stratification, multimodal machine learning, and explainability in healthcare. It highlights limitations in current approaches and establishes the need for the proposed framework.

2.1. Lung disease risk stratification

Lung diseases, including pneumonia, chronic obstructive pulmonary disease (COPD), and lung cancer, continue to pose a major global health burden, contributing significantly to mortality and healthcare costs worldwide (Chen et al., 2023). Traditional risk stratification tools, such as the CURB-65 for pneumonia and the BODE index for COPD, have long been used to guide clinical decision-making (Barradell, Bourne,

Alkhatlan, Larkin, & Singh, 2022). These tools rely on a predefined set of clinical features, often failing to capture the complexity and heterogeneity of lung disease presentations. While they offer valuable insights for specific diseases, their applicability is limited when dealing with multifaceted conditions or comorbidities common in lung disease patients (dos Santos et al., 2022).

In recent years, the advent of machine learning has introduced a new paradigm for lung disease risk stratification (Khanna et al., 2022). Models trained on large-scale datasets have demonstrated the potential to surpass traditional tools by identifying subtle patterns and incorporating a broader range of features (Juge et al., 2022; Khanna et al., 2022). However, many of these models are unimodal, focusing solely on clinical or imaging data, and lack the flexibility to adapt dynamically to patient-specific contexts (Jiang & Veeraraghavan, 2022). This limitation hinders their ability to provide robust and personalized risk predictions, underscoring the need for frameworks that can effectively integrate and align multimodal data sources.

2.2. Multimodal machine learning in healthcare

The integration of structured clinical data, such as electronic health records (EHRs), with unstructured imaging data has emerged as a powerful approach for enhancing predictive accuracy in healthcare (Ros & Viejo, 2024; Hong-Yu Zhou et al., 2023; Zhou et al., 2024). By leveraging diverse data modalities, multimodal machine learning models can capture complementary information, providing a more holistic view of patient health. Early fusion techniques, which combine data modalities at the input stage through concatenation or averaging, have been widely explored (Kazdal, Hofman, Christopoulos, Ilić, Stenzinger, & Hofman, 2022; Hong-Yu Zhou et al., 2023). However, these methods often fail to account for the complex interdependencies between modalities, limiting their effectiveness in capturing nuanced interactions (Kazdal et al., 2022; Kumar et al., 2022).

Recent advances in attention-based models, particularly those leveraging Transformer architectures, have addressed some of these limitations by enabling more sophisticated fusion mechanisms (Mezina & Burget, 2024; Xie, Fang, Yang, Yu, & Li, 2023). These models dynamically weigh the contributions of different modalities, enhancing their ability to capture complex relationships. Despite these advancements, many existing frameworks rely on static fusion mechanisms that overlook the dynamic nature of clinical data. Furthermore, most approaches fail to prioritize modality-specific contributions based on the patient's unique context, which is essential for achieving personalized healthcare solutions (Barradell et al., 2022; Chen et al., 2023).

2.3. Explainability in multimodal models

Explainability is a critical requirement for deploying machine learning models in clinical settings, where trust and transparency are paramount. Techniques such as SHAP (SHapley Additive exPlanations) (Wani, Kumar, & Bedi, 2024) and Grad-CAM (Gradient-weighted Class Activation Mapping) have gained prominence for interpreting structured and imaging data, respectively. These methods provide localized insights into feature importance or highlight regions of interest within medical images, offering clinicians valuable interpretative tools. However, they are typically applied in isolation for each modality, leaving clinicians to manually synthesize insights from separate explainability outputs (prasad Koyyada & Singh, 2023).

The integration of explainability within multimodal frameworks remains an underexplored area (Kumar, Shvetsov, & Alsamhi, 2024). Existing methods often fail to provide a cohesive interpretability framework that accounts for the interplay between structured clinical features and imaging findings (Yang, Ye, & Xia, 2022). This gap limits the utility of multimodal models in real-world clinical decision-making, where the ability to derive actionable insights across modalities is crucial. Addressing these limitations requires novel approaches that integrate explainability seamlessly into the modeling process, enabling clinicians to make informed decisions based on comprehensive and interpretable predictions (Sun, Akman, & Schuller, 2024).

2.4. Research gaps

While multimodal machine learning has demonstrated potential in lung disease risk stratification, existing approaches suffer from critical limitations that hinder their clinical applicability. Most prior studies employ static fusion techniques, where clinical and imaging features are combined without dynamically adapting to individual patient characteristics. This rigid integration fails to capture context-dependent modality relevance, leading to suboptimal predictions. Unlike these methods, CAMAF introduces a Context-Gated Attention (CGA) mechanism, which adaptively reweights modality contributions based on patient-specific data, ensuring a more precise and personalized risk assessment. Additionally, existing models often apply explainability techniques in isolation, either focusing on structured clinical data or imaging data separately, limiting their holistic interpretability. In contrast, CAMAF integrates SHAP for clinical feature attribution and Grad-CAM for visual explanations, providing a cohesive multimodal interpretability framework that enhances clinical trust. Furthermore, while prior approaches primarily evaluate model accuracy, they fail to benchmark against traditional clinical risk scoring systems (e.g., CURB-65, BODE Index). CAMAF directly compares with these scoring systems, demonstrating superior performance, thereby establishing its real-world relevance. These limitations highlight the need for an advanced multimodal risk stratification framework that combines dynamic alignment, robust fusion strategies, and unified explainability, a gap that CAMAF effectively addresses.

3. Methodology

The proposed Context-Aware Multimodal Alignment Framework (CAMAF) is a novel approach designed to integrate structured clinical data and unstructured imaging data for dynamic and explainable stratification of risk of lung disease. The process involves multiple stages, as shown in Fig. 1. Clinical data is processed using the TabTransformer (Yang et al., 2024), while imaging data is analyzed using the Vision Transformer (ViT) (Goceri, 2024), ensuring optimal feature extraction from both modalities. A Temporal Fusion Transformer (TFT) (Wang et al., 2024) generates a patient-specific context vector by combining static and temporal clinical features. This vector guides the Context-Gated Attention (CGA) mechanism to dynamically adjust the contributions of each modality, aligning them effectively. The fusion layer concatenates the weighted embeddings, modeling interdependencies through an interaction layer, while the classification layer leverages a Transformer Encoder to produce robust and accurate risk predictions. CAMAF's design emphasizes adaptability, interpretability, and the ability to model complex relationships between heterogeneous data modalities, making it well-suited for lung disease risk assessment.

3.1. Feature embedding

The feature embedding stage processes structured clinical data and unstructured imaging data to generate meaningful representations for downstream tasks. These embeddings are produced using the TabTransformer for clinical data and the Vision Transformer (ViT) for imaging data.

Clinical data embedding. Structured clinical data is represented as a matrix $\mathbf{X}_c \in \mathbb{R}^{N \times D_c}$, where N is the number of samples and D_c is the number of clinical features. TabTransformer encodes the data as follows:

$$\mathbf{E}_c = \text{TabTransformer}(\mathbf{X}_c), \quad (1)$$

where $\mathbf{E}_c \in \mathbb{R}^{N \times H_c}$ is the clinical embedding matrix, and H_c is the embedding dimensionality. TabTransformer leverages attention mechanisms to preserve feature-level interpretability and capture inter-feature dependencies.

Imaging data embedding. Imaging data, such as chest X-rays, is represented as a set of pixel intensity matrices $\mathbf{X}_i \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote the image height, width, and number of channels, respectively. The Vision Transformer processes this data by dividing it into patches of size $P \times P$, which are flattened into patch embeddings:

$$\mathbf{P}_i = \text{PatchEmbedding}(\mathbf{X}_i), \quad (2)$$

where $\mathbf{P}_i \in \mathbb{R}^{N_p \times D_p}$, $N_p = \frac{H \times W}{P^2}$ is the number of patches, and D_p is the dimensionality of each patch embedding. These patch embeddings are passed through the Vision Transformer:

$$\mathbf{E}_i = \text{ViT}(\mathbf{P}_i), \quad (3)$$

where $\mathbf{E}_i \in \mathbb{R}^{N_p \times H_i}$ is the imaging embedding matrix, and H_i is the embedding dimensionality.

Unified representations. The embeddings from the TabTransformer and ViT are used as inputs to the subsequent components of the framework:

$$\mathbf{E}_c, \mathbf{E}_i \rightarrow \text{Context Encoder and Dynamic Alignment Layer}. \quad (4)$$

These embeddings capture complementary structured and unstructured information, forming a robust foundation for the context-aware alignment and fusion processes.

3.2. Context encoder

The context encoder generates a patient-specific context vector by leveraging static demographic features and temporal trends in clinical data. This vector guides the alignment of clinical and imaging embeddings in subsequent layers. The context encoder takes two types of input:

- Static features: $\mathbf{X}_c^s \in \mathbb{R}^{N \times D_s}$, where N is the number of samples and D_s is the dimensionality of static features (e.g., age, gender, BMI).
- Temporal features: $\mathbf{X}_c^t \in \mathbb{R}^{N \times T \times D_t}$, where T is the number of time steps and D_t is the dimensionality of temporal features (e.g., lab results over time).

Static feature encoding. Static features are transformed using a fully connected layer:

$$\mathbf{E}_c^s = \sigma(\mathbf{W}_s \mathbf{X}_c^s + \mathbf{b}_s), \quad (5)$$

where $\mathbf{W}_s \in \mathbb{R}^{D_s \times H_c}$, $\mathbf{b}_s \in \mathbb{R}^{H_c}$ are trainable parameters, H_c is the embedding size, and σ is an activation function (e.g., ReLU).

Temporal feature encoding. Temporal features are processed using a multi-head self-attention mechanism to capture dependencies over time:

$$\mathbf{E}_c^t = \text{MultiHeadAttention}(\mathbf{X}_c^t), \quad (6)$$

where $\mathbf{E}_c^t \in \mathbb{R}^{N \times T \times H_c}$ represents the attended temporal embeddings.

Temporal fusion. The attended temporal embeddings are combined with the static embeddings to produce a fused representation:

$$\mathbf{E}_{c,\text{fused}} = \mathbf{E}_c^t + \mathbf{E}_c^s, \quad (7)$$

where $\mathbf{E}_{c,\text{fused}} \in \mathbb{R}^{N \times T \times H_c}$ integrates both static and temporal information.

Context vector generation. The fused representation is aggregated across the temporal dimension to generate a patient-specific context vector:

$$\mathbf{C}_c = \text{Aggregate}(\mathbf{E}_{c,\text{fused}}), \quad (8)$$

where the aggregation function (e.g., mean or attention pooling) reduces the temporal dimension, resulting in $\mathbf{C}_c \in \mathbb{R}^{N \times H_c}$.

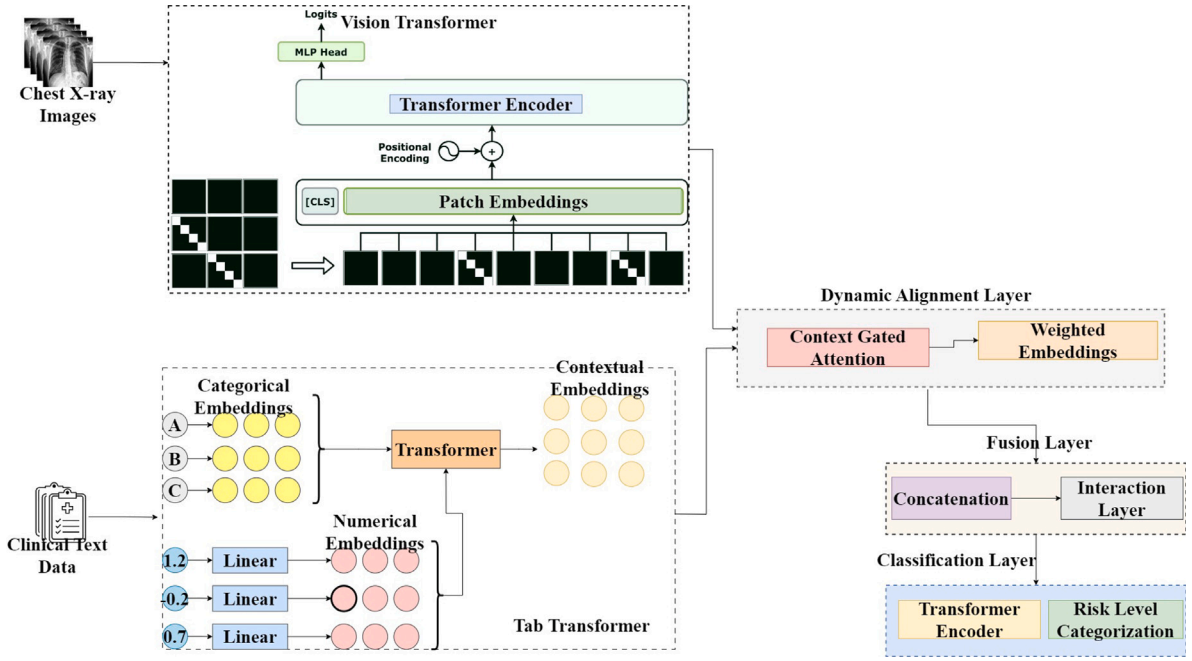


Fig. 1. An overview of the proposed Context-Aware Multimodal Alignment Framework (CMAAF).

Output. The context vector C_c is used as an input to the dynamic alignment layer:

$$C_c \rightarrow \text{Dynamic Alignment Layer.} \quad (9)$$

This vector encapsulates both static and temporal information, enabling personalized alignment of clinical and imaging embeddings.

3.3. Dynamic alignment layer

The dynamic alignment layer is responsible for dynamically weighting the clinical and imaging embeddings based on the patient-specific context vector C_c generated by the context encoder. This mechanism ensures that the contributions of each modality are adjusted according to the individual characteristics of the patient. The inputs to the dynamic alignment layer are:

- Clinical embeddings: $E_c \in \mathbb{R}^{N \times H_c}$, derived from the TabTransformer.
- Imaging embeddings: $E_i \in \mathbb{R}^{N \times H_i}$, derived from the Vision Transformer.
- Context vector: $C_c \in \mathbb{R}^{N \times H_c}$, generated by the context encoder.

Context-Gated Attention. The proposed Context-Gated Attention (CGA) mechanism dynamically computes weights for the clinical and imaging embeddings using the context vector C_c . The gating weights are calculated as follows:

$$G_c = \sigma(Wc1Cc + bc1), \quad G_i = 1 - G_c, \quad (10)$$

where $Wc1 \in \mathbb{R}^{H_c \times H_c}$ and $bc1 \in \mathbb{R}^{H_c}$ are trainable parameters, and σ is the sigmoid activation function.

Weighted embeddings. Using the gating weights, the clinical and imaging embeddings are dynamically weighted:

$$E_{c, \text{weighted}} = G_c \odot E_c, \quad E_{i, \text{weighted}} = G_i \odot E_i, \quad (11)$$

where $\odot$ represents element-wise multiplication. The outputs of the dynamic alignment layer are the weighted clinical and imaging embeddings:

$$E_{\text{aligned}} = E_{c, \text{weighted}} \parallel E_{i, \text{weighted}}. \quad (12)$$

These aligned embeddings are subsequently used in the fusion layer for inter-modality interaction modeling. The CGA mechanism ensures that the contributions of clinical and imaging data are adaptively modulated based on patient-specific contexts, enhancing the relevance and robustness of the model's predictions.

3.4. Fusion layer

The fusion layer combines the dynamically aligned clinical and imaging embeddings to model interdependencies between the two modalities effectively. This step ensures that the complementary information from clinical and imaging data is integrated into a unified representation for downstream tasks. The inputs to the fusion layer are the aligned embeddings generated by the dynamic alignment layer:

- Weighted clinical embeddings: $E_{c, \text{weighted}} \in \mathbb{R}^{N \times H_c}$.
- Weighted imaging embeddings: $E_{i, \text{weighted}} \in \mathbb{R}^{N \times H_i}$.

Concatenation. The weighted embeddings from the clinical and imaging modalities are concatenated to form a unified embedding:

$$E_{\text{fused}} = [E_{c, \text{weighted}} \parallel E_{i, \text{weighted}}], \quad (13)$$

where $E_{\text{fused}} \in \mathbb{R}^{N \times (H_c + H_i)}$ represents the concatenated embedding that combines information from both modalities.

Interaction layer. The fused embedding is passed through a fully connected interaction layer to model interactions between the clinical and imaging features:

$$E_{\text{interaction}} = \sigma(W_f E_{\text{fused}} + b_f), \quad (14)$$

where $W_f \in \mathbb{R}^{(H_c + H_i) \times H_f}$ and $b_f \in \mathbb{R}^{H_f}$ are trainable parameters, H_f is the hidden dimensionality, and σ is the activation function (e.g., ReLU). The output of the fusion layer is the interaction embedding:

$$E_{\text{output}} = E_{\text{interaction}} \in \mathbb{R}^{N \times H_f}. \quad (15)$$

This embedding serves as the unified representation for both modalities and is forwarded to the prediction head for classification or regression tasks. The fusion layer effectively captures the synergistic relationships between clinical and imaging data, enhancing the model's ability to make robust predictions.

3.5. Classification layer

The classification layer is the final component of the framework, responsible for predicting discrete risk levels (*low*, *moderate*, and *high*) based on the unified embeddings produced by the fusion layer. These risk levels are derived by stratifying patients according to clinical and imaging features indicative of disease severity, which are determined using domain-specific thresholds and clinical guidelines. The input to the classification layer is the unified embedding from the fusion layer:

$$\mathbf{E}_{\text{output}} \in \mathbb{R}^{N \times H_f}, \quad (16)$$

where N is the batch size, and H_f is the embedding dimensionality.

Transformer encoder. The Transformer Encoder processes the fused embeddings to capture complex relationships among the features. It consists of two main components:

- **Multi-Head Self-Attention:** Captures dependencies across all features in the embedding:

$$\mathbf{Z} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V}, \quad (17)$$

where $\mathbf{Q} = \mathbf{E}_{\text{output}} \mathbf{W}_Q$, $\mathbf{K} = \mathbf{E}_{\text{output}} \mathbf{W}_K$, and $\mathbf{V} = \mathbf{E}_{\text{output}} \mathbf{W}_V$ are the query, key, and value matrices, respectively. Here, $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{H_f \times d_k}$ are trainable projection matrices, and d_k is the dimensionality of each key vector.

- **Feedforward Network:** Applies a position-wise fully connected network with a non-linear activation:

$$\mathbf{F} = \text{ReLU}(\mathbf{Z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2, \quad (18)$$

where $\mathbf{W}_1 \in \mathbb{R}^{H_f \times H_{ff}}$, $\mathbf{W}_2 \in \mathbb{R}^{H_{ff} \times H_f}$ are the trainable weight matrices, $\mathbf{b}_1, \mathbf{b}_2$ are the biases, and H_{ff} is the hidden dimensionality of the feedforward layer.

Output layer. The final output is generated by applying a fully connected layer with a softmax activation function for the classification of risk levels:

$$\mathbf{y}_{\text{pred}} = \text{softmax}(\mathbf{F}\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}), \quad (19)$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{H_f \times K}$, $\mathbf{b}_{\text{out}} \in \mathbb{R}^K$, and K is the number of risk levels (*low*, *moderate*, *high*). The classification layer produces the final predictions:

$$\mathbf{y}_{\text{pred}} \in \mathbb{R}^{N \times K}, \quad (20)$$

representing the probabilities of the input samples belonging to each risk category.

Risk level categorization. The categorization of patients into *low*, *moderate*, and *high* risk is determined using the following quantifiable criteria:

- **Clinical thresholds:** Risk levels are defined based on clinical markers. For example:
 - *Low risk:* Oxygen saturation (SpO_2) > 94%, respiratory rate (RR) between 12 and 20 breaths/min, and lactate levels < 2 mmol/L.
 - *Moderate risk:* SpO_2 between 88%–94%, RR between 20 and 30 breaths/min, and lactate levels between 2–4 mmol/L.
 - *High risk:* SpO_2 < 88%, RR > 30 breaths/min, or lactate levels > 4 mmol/L, indicating severe respiratory distress or metabolic derangement.
- **Imaging scores:** Radiological findings are assessed based on scoring systems for abnormalities such as lung opacities and pleural effusions:

- *Low risk:* Minimal or no findings, e.g., opacity score < 2 (on a 0–8 scale) or no pleural effusion.
- *Moderate risk:* Moderate findings, e.g., opacity score between 2–5 or small to moderate pleural effusion.
- *High risk:* Severe abnormalities, e.g., opacity score > 5 or large pleural effusion.

- **Weighted integration:** The Context-Gated Attention mechanism dynamically adjusts the contributions of clinical and imaging data based on patient-specific context. For example, patients with borderline clinical parameters but severe radiological abnormalities are shifted toward a higher risk category. This is achieved by assigning modality-specific weights (e.g., clinical weight: 0.6, imaging weight: 0.4) that adapt dynamically based on the input features.

The combination of these criteria ensures that each patient's risk category is assigned based on a comprehensive evaluation of clinical and imaging features, making the predictions interpretable and clinically actionable. The pseudocode of the proposed CAMAF framework is shown in Algorithm 1.

Algorithm 1 Context-Aware Multimodal Alignment Framework (CAMAF)

Require: Clinical data $\mathbf{X}_c \in \mathbb{R}^{N \times D_c}$, Imaging data $\mathbf{X}_i \in \mathbb{R}^{H \times W \times C}$

Ensure: Predicted probabilities $\mathbf{y}_{\text{pred}}$

- 1: **Initialize:** Parameters for TabTransformer, Vision Transformer (ViT), Temporal Fusion Transformer (TFT), and Transformer Encoder.
 - 2: **Stage 1: Feature Embedding**
 - 3: $\mathbf{E}_c \leftarrow \text{TabTransformer}(\mathbf{X}_c)$ *Extract clinical embeddings.*
 - 4: $\mathbf{P}_i \leftarrow \text{PatchEmbedding}(\mathbf{X}_i)$ *Convert imaging data to patch embeddings.*
 - 5: $\mathbf{E}_i \leftarrow \text{ViT}(\mathbf{P}_i)$ *Extract imaging embeddings.*
 - 6: **Stage 2: Context Encoder**
 - 7: $\mathbf{E}_c^s \leftarrow \sigma(\mathbf{W}_s \mathbf{X}_c^s + \mathbf{b}_s)$ *Encode static clinical features.*
 - 8: $\mathbf{E}_c^t \leftarrow \text{MultiHeadAttention}(\mathbf{X}_c^t)$ *Encode temporal clinical features.*
 - 9: $\mathbf{E}_c^{\text{fused}} \leftarrow \mathbf{E}_c^s + \mathbf{E}_c^t$ *Fuse static and temporal features.*
 - 10: $\mathbf{C}_c \leftarrow \text{Aggregate}(\mathbf{E}_c^{\text{fused}})$ *Generate patient-specific context vector.*
 - 11: **Stage 3: Dynamic Alignment Layer**
 - 12: $G_c \leftarrow \sigma(\mathbf{W}_c \mathbf{C}_c + \mathbf{b}_c)$ *Compute clinical gating weight.*
 - 13: $G_i \leftarrow 1 - G_c$ *Compute imaging gating weight.*
 - 14: $\mathbf{E}_c^{\text{weighted}} \leftarrow G_c \odot \mathbf{E}_c$ *Weight clinical embeddings.*
 - 15: $\mathbf{E}_i^{\text{weighted}} \leftarrow G_i \odot \mathbf{E}_i$ *Weight imaging embeddings.*
 - 16: **Stage 4: Fusion Layer**
 - 17: $\mathbf{E}_{\text{fused}} \leftarrow [\mathbf{E}_c^{\text{weighted}}; \mathbf{E}_i^{\text{weighted}}]$ *Concatenate embeddings.*
 - 18: $\mathbf{E}_{\text{interaction}} \leftarrow \sigma(\mathbf{W}_f \mathbf{E}_{\text{fused}} + \mathbf{b}_f)$ *Model interdependencies.*
 - 19: **Stage 5: Classification Layer**
 - 20: $\mathbf{Z} \leftarrow \text{MultiHeadAttention}(\mathbf{E}_{\text{interaction}})$ *Apply self-attention.*
 - 21: $\mathbf{F} \leftarrow \text{ReLU}(\mathbf{Z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$ *Apply feedforward network.*
 - 22: $\mathbf{y}_{\text{pred}} \leftarrow \text{softmax}(\mathbf{F}\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}})$ *Generate predictions.*
 - 23: **return** $\mathbf{y}_{\text{pred}}$
-

4. Experimental evaluation

This section presents the experimental setup, evaluation metrics, and a detailed analysis of the results obtained by the proposed CAMAF.

4.1. Datasets

This study utilizes two prominent datasets for evaluating the proposed framework: the CheXpert dataset (Nupnau, Kattan, & Reynolds, 2024) and the MIMIC-IV dataset (Sadeghi, Hempel, Rodemund, & Kirsten, 2024). These datasets provide complementary information, with CheXpert focusing on imaging data and MIMIC-IV encompassing detailed clinical records.

Table 1
Statistics of the CheXpert and MIMIC-IV datasets.

Dataset	Modality	Attribute	Value
CheXpert	Imaging	Training Images	224,316
		Validation Images	234
		Testing Images	500
MIMIC-IV	Clinical	Unique Patients	364,627
		Hospitalizations	546,028
		ICU Stays	94,458
		Data Modules	hosp, icu

CheXpert dataset. The CheXpert dataset is a large-scale chest X-ray dataset released by the Stanford Machine Learning Group. It comprises 224,316 high-quality X-ray images from 65,240 patients in the training set, 234 images from 200 patients in the validation set, and a testing set with images for 500 patients. The testing data is maintained by the organizers for final evaluation and is not publicly released. The dataset was annotated to identify the presence of 14 observations commonly found in radiology reports, such as Cardiomegaly, Edema, Consolidation, Atelectasis, and Pleural Effusion. The training annotations were generated using an automated labeler, while the validation images were manually annotated by three board-certified radiologists, and the testing images were reviewed by a consensus of five board-certified radiologists. The average resolution of CheXpert images is 2828×2320 pixels, significantly larger than typical datasets like ImageNet.

MIMIC-IV dataset. The Medical Information Mart for Intensive Care (MIMIC-IV) is a comprehensive and deidentified dataset of electronic health records (EHRs) collected from patients admitted to the emergency department or intensive care unit (ICU) at the Beth Israel Deaconess Medical Center. MIMIC-IV contains data for 364,627 unique individuals, representing 546,028 hospitalizations and 94,458 unique ICU stays. This modular dataset is divided into two main components: the *hosp* module, which contains hospital-wide EHR data, and the *icu* module, which provides detailed clinical information for ICU patients. MIMIC-IV supports a broad range of applications, from algorithm development to healthcare insights, and is recognized for its contribution to improving patient care while ensuring privacy.

As shown in Table 1, the CheXpert dataset provides extensive imaging data with annotations for common radiological observations, while the MIMIC-IV dataset offers detailed clinical and temporal data, enabling a comprehensive evaluation of the proposed framework.

4.2. Experimental setup

The experiments were conducted to evaluate the performance of the proposed Context-Aware Multimodal Alignment Framework (CAMAF) for lung disease risk stratification. The implementation was carried out using the PyTorch framework, and the training was performed on a system equipped with an NVIDIA Tesla V100 GPU, 64 GB RAM, and an Intel Xeon CPU. The TabTransformer and Vision Transformer (ViT) were pre-trained on large-scale datasets and fine-tuned for the respective clinical and imaging modalities.

To address class imbalance, we applied a combination of class-weighting, focal loss, and data augmentation techniques. Class weights were assigned inversely proportional to the frequency of each pathology to reduce bias toward majority classes. Additionally, focal loss was employed to prioritize learning from underrepresented pathologies like Pneumonia and Pneumothorax. For imaging data, oversampling of rare conditions was performed using geometric transformations such as rotation, flipping, and contrast enhancement, ensuring balanced exposure during training. The CheXpert and MIMIC-IV datasets were preprocessed by normalizing pixel intensities for imaging data and handling missing values in clinical data using mean imputation. The train-validation-test split ratio was set to 70:15:15, ensuring robust evaluation and mitigating potential biases introduced by dataset imbalances.

4.3. Evaluation metrics

The performance of the proposed Context-Aware Multimodal Alignment Framework (CAMAF) is evaluated using a combination of standard and advanced metrics to assess predictive accuracy, robustness, and explainability.

Classification metrics include Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Accuracy measures the overall correctness of predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (21)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. Precision evaluates the proportion of true positives among predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (22)$$

Recall (Sensitivity) measures the ability to identify true positives:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (23)$$

The F1-Score balances precision and recall:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (24)$$

The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) evaluates the model's ability to distinguish between classes by plotting true positive rate (TPR) against false positive rate (FPR) at different thresholds:

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR}. \quad (25)$$

Advanced metrics include Expected Calibration Error (ECE), which measures the alignment between predicted probabilities and observed frequencies:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} \cdot |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (26)$$

where B_m is the m th probability bin, $\text{acc}(B_m)$ is the accuracy, $\text{conf}(B_m)$ is the confidence, and N is the total number of samples. Cohen's Kappa assesses agreement between predictions and true labels, accounting for chance agreement:

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \quad (27)$$

where P_o is the observed agreement, and P_e is the expected agreement by chance.

4.4. Baselines

To evaluate the effectiveness of the proposed Context-Aware Multimodal Alignment Framework (CAMAF), five baseline models were implemented. These include: (1) a clinical-only model using TabTransformer to process structured clinical data exclusively, (2) an imaging-only model using Vision Transformer (ViT) trained on chest X-ray data, (3) an early fusion model that concatenates clinical and imaging features (Vidyasri & Saravanan, 2023), followed by a multi-layer perceptron (MLP) for classification, (Thavasimuthu, Hanumanthakari, Sekar, and Kirubakaran (2024)) (4) a late fusion model (Ibrahim et al., 2024) combining predictions from independent clinical and imaging models using weighted averaging, and (5) a multimodal attention-based model (Narmadha & Gobalakrishnan, 2025) employing a standard attention mechanism to integrate clinical and imaging embeddings. These baselines were chosen to benchmark unimodal performance, evaluate traditional multimodal fusion techniques, and compare against other attention-based methods. This setup ensures a comprehensive assessment of CAMAF's ability to leverage context-aware alignment and dynamic fusion for improved prediction accuracy and interpretability.

Table 2

Hyperparameter tuning results for CAMAF. The best-performing configuration is highlighted.

Hyperparameter	Explored range	Optimal value	Impact on performance
Learning Rate	1×10^{-5} to 1×10^{-3}	1×10^{-4}	Higher rates led to instability, lower slowed convergence
Batch Size	16, 32, 64	32	Balanced performance and memory efficiency
Transformer Layers	2, 4, 6	4	More layers led to overfitting without gains
Attention Heads	4, 8, 12	8	Increasing beyond 8 had minimal effect
Dropout Rate	0.1, 0.3, 0.5	0.3	Higher dropout improved generalization
Weight Decay	10^{-6} to 10^{-3}	10^{-5}	Helped prevent overfitting
Optimizer	Adam, AdamW, SGD	AdamW	Best convergence and stability

Table 3

Performance comparison of CAMAF with baseline models.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Clinical-Only (TabTransformer)	0.8421	0.8324	0.8352	0.8338	0.8923
Imaging-Only (ViT)	0.8517	0.8432	0.8450	0.8441	0.9015
Early Fusion (MLP)	0.8773	0.8637	0.8685	0.8661	0.9148
Late Fusion (Weighted Average)	0.8625	0.8530	0.8574	0.8552	0.9102
Attention-Based Fusion	0.9023	0.8905	0.8937	0.8921	0.9234
CAMAF (Proposed)	0.9278	0.9142	0.9183	0.9162	0.9457

4.5. Hyperparameter tuning and selection

The hyperparameters for CAMAF were optimized using a grid search strategy to achieve the best trade-off between performance and computational efficiency. The grid search explored different configurations for key parameters, including learning rate, batch size, number of Transformer encoder layers, and attention heads. The optimal values were selected based on the highest validation AUC-ROC score while maintaining a balanced trade-off between precision, recall, and computational cost. The explored hyperparameter ranges and the selected optimal values are summarized in Table 2. The learning rate was varied from 1×10^{-5} to 1×10^{-3} , with the best results obtained at 1×10^{-4} . The batch size was tested between 16 and 64, with 32 yielding the highest accuracy without excessive memory overhead. Increasing the number of attention heads beyond 8 showed diminishing returns, while four Transformer encoder layers provided optimal contextual learning without overfitting.

These results indicate that CAMAF benefits from a moderately aggressive learning rate, a sufficiently large batch size for stable gradient updates, and a carefully selected number of Transformer layers and attention heads. The optimized configuration provides a strong balance between generalization and computational efficiency, ensuring robust performance in lung disease risk stratification.

4.6. Results and discussion

4.6.1. Evaluation with baselines

The proposed Context-Aware Multimodal Alignment Framework (CAMAF) was evaluated against five baselines using multiple performance metrics: Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Table 3 summarizes the results for each model, demonstrating the superior performance of CAMAF across all metrics.

The results in Table 3 highlight the effectiveness of CAMAF in achieving an accuracy of 0.9278, significantly outperforming the unimodal models (Clinical-Only and Imaging-Only) and traditional fusion methods (Early Fusion and Late Fusion). CAMAF also surpasses the attention-based fusion baseline, showcasing the impact of its context-aware dynamic alignment and multimodal fusion mechanisms. Across all metrics, CAMAF demonstrates a balanced improvement in Precision, Recall, and F1-Score, indicating robust performance in handling both true positives and false negatives. The higher AUC-ROC score (0.9457) further validates its superior discriminative capability, emphasizing its ability to effectively integrate clinical and imaging modalities for lung disease risk stratification.

4.6.2. Sensitivity analysis

A sensitivity analysis was conducted to evaluate the impact of key hyperparameters on the performance of the proposed CAMAF framework for lung disease risk stratification. The hyperparameters analyzed included the learning rate, batch size, and the number of attention heads in the Transformer Encoder. Metrics such as Accuracy and AUC-ROC were used to assess the model's predictive performance across different configurations.

The results, presented in Fig. 2, demonstrate that CAMAF is robust to changes in hyperparameter values within a reasonable range. However, significant performance degradation was observed when the learning rate exceeded 10^{-3} or the batch size was reduced below 16. Additionally, while increasing the number of attention heads improved performance up to eight heads, further increases yielded diminishing returns. These findings highlight the importance of choosing balanced configurations to ensure stable and optimal performance in clinical risk stratification scenarios.

4.6.3. Ablation study

To evaluate the contributions of key components in the Context-Aware Multimodal Alignment Framework (CAMAF) for lung disease risk stratification, an ablation study was conducted. The analysis examined the effects of removing or modifying critical components, including the Context Encoder, Context-Gated Attention (CGA), and the Transformer Encoder in the classification layer. Variants using simpler fusion mechanisms, such as early and late fusion, were also considered. Table 4 presents the performance metrics across Accuracy, Precision, Recall, F1-Score, and AUC-ROC, highlighting each component's impact.

The results in Table 4 demonstrate the critical role of advanced components in CAMAF for accurate lung disease stratification. The removal of the Context Encoder caused a significant drop in performance (Accuracy: 0.8905, AUC-ROC: 0.9173), highlighting the importance of integrating static and temporal clinical data for patient-specific alignment. Substituting the Context-Gated Attention (CGA) mechanism with standard attention led to a decline in accuracy (0.9032) and AUC-ROC (0.9285), emphasizing the value of dynamic modality weighting for effective integration of clinical and imaging data. The absence of the Transformer Encoder in the classification layer reduced AUC-ROC to 0.9067, indicating its pivotal role in capturing interdependencies between fused features. Variants using early or late fusion methods performed significantly worse across all metrics, as they failed to fully model the complementary relationships between structured and unstructured data. These findings validate the design choices of CAMAF, particularly the inclusion of the Context Encoder, CGA, and Transformer Encoder, in achieving robust and interpretable stratification of lung disease risk.

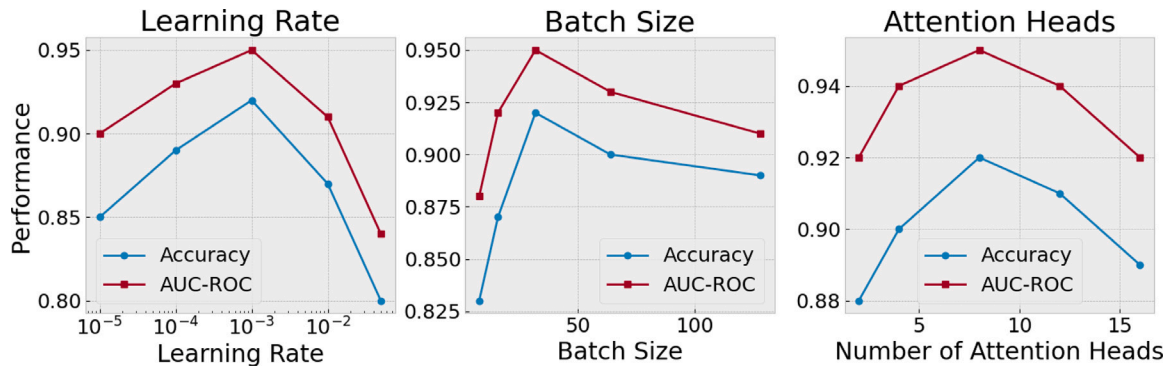


Fig. 2. Sensitivity analysis of CAMAF under varying hyperparameter configurations.

Table 4

Ablation study results for CAMAF, highlighting the importance of each component for lung disease risk stratification.

Model variant	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Full CAMAF	0.9278	0.9142	0.9183	0.9162	0.9457
w/o Context Encoder	0.8905	0.8724	0.8779	0.8751	0.9173
w/o CGA (Standard Attention)	0.9032	0.8911	0.8954	0.8932	0.9285
w/o Transformer Encoder	0.8731	0.8579	0.8635	0.8606	0.9067
Early Fusion Only	0.8746	0.8607	0.8669	0.8638	0.9092
Late Fusion Only	0.8664	0.8521	0.8576	0.8548	0.9041

Table 5

Statistical significance of CAMAF improvements over baselines (p-values).

Comparison model	Accuracy (p-value)	F1-Score (p-value)	AUC-ROC (p-value)
Clinical-Only (TabTransformer)	< 0.001	< 0.001	< 0.001
Imaging-Only (ViT)	< 0.001	< 0.001	< 0.001
Early Fusion (MLP)	0.002	0.004	0.003
Late Fusion (Weighted Average)	0.005	0.008	0.006
Attention-Based Fusion	0.021	0.015	0.018

4.6.4. Statistical analysis

To validate the significance of the observed performance differences between CAMAF and the baseline models, a statistical analysis was conducted. Paired t-tests were performed on key evaluation metrics (Accuracy, F1-Score, and AUC-ROC) to determine whether the improvements achieved by CAMAF were statistically significant. The p-values for these tests are summarized in Table 5.

The statistical analysis in Table 5 confirms that CAMAF significantly outperforms all baseline models across Accuracy, F1-Score, and AUC-ROC metrics. For the unimodal models (Clinical-Only and Imaging-Only), the p-values were consistently below 0.001, demonstrating highly significant improvements. Similarly, CAMAF showed statistically significant advantages over traditional fusion approaches (Early Fusion and Late Fusion), with p-values ranging between 0.002 and 0.008. While CAMAF also outperformed the attention-based fusion baseline, the differences were less pronounced but still statistically significant, with p-values below 0.05. These results validate the efficacy of CAMAF's context-aware dynamic alignment and multimodal fusion in achieving superior predictive performance with high statistical confidence.

4.6.5. Robustness analysis

The robustness of the proposed Context-Aware Multimodal Alignment Framework (CAMAF) was evaluated under varying conditions, including different levels of input data noise and reduced data availability. These scenarios reflect real-world challenges in lung disease risk stratification, where clinical and imaging data may be noisy, incomplete, or imbalanced. Two metrics were used for this analysis: Expected Calibration Error (ECE), which measures the alignment between predicted probabilities and observed outcomes, and Cohen's Kappa, which assesses agreement between predictions and ground truth, accounting

for chance. The results of the robustness analysis are summarized in Table 6.

Table 6 illustrates CAMAF's superior robustness compared to the best baseline (Attention-Based Fusion) across all tested conditions. Under no-noise conditions, CAMAF achieves an ECE of 0.028 and a Kappa score of 0.921, demonstrating highly calibrated predictions and strong agreement with the ground truth. When Gaussian noise is introduced to the input data, CAMAF's performance degrades gracefully, maintaining a Kappa score of 0.889 under high noise, compared to 0.821 for the baseline. This indicates CAMAF's ability to handle noisy imaging and clinical data commonly encountered in healthcare settings. Similarly, CAMAF outperforms the baseline under reduced data availability. With only 50% of the data, CAMAF achieves an ECE of 0.031 and a Kappa score of 0.904, reflecting its resilience in scenarios with limited training samples. The robustness of CAMAF can be attributed to its context-aware dynamic alignment and multimodal fusion mechanisms, which effectively integrate heterogeneous data sources even under suboptimal conditions. These results underscore CAMAF's reliability for lung disease risk stratification in real-world clinical applications.

4.6.6. Class-wise performance analysis

To evaluate the contributions of individual pathologies and combinations of pathologies to lung disease risk stratification, a class-wise analysis was conducted using the CheXpert dataset. The analysis measured the performance of CAMAF for each of the 14 labeled observations, as well as its ability to stratify patients into specific risk categories (*low*, *moderate*, *high*). The risk categories were defined based on quantified clinical thresholds and imaging scores:

- **Low Risk:** $\text{SpO}_2 > 94\%$, respiratory rate (RR) between 12 and 20 breaths/min, and opacity score < 2 with no significant radiological findings.

Table 6

Robustness analysis results for CAMAF and baselines under varying conditions relevant to lung disease stratification.

Condition	Model	ECE (Lower is better)	Kappa (Higher is better)
No noise (Baseline)	CAMAF	0.028	0.921
	Best Baseline (Attention-Based)	0.051	0.874
Low noise (Gaussian)	CAMAF	0.035	0.910
	Best Baseline (Attention-Based)	0.062	0.852
High noise (Gaussian)	CAMAF	0.049	0.889
	Best Baseline (Attention-Based)	0.079	0.821
50% data availability	CAMAF	0.031	0.904
	Best Baseline (Attention-Based)	0.059	0.847
75% data availability	CAMAF	0.029	0.915
	Best Baseline (Attention-Based)	0.054	0.866

Table 7

Class-wise and risk category performance metrics of CAMAF for lung disease risk stratification.

Pathology/Risk category	Accuracy	Precision	Recall	F1-Score	AUC-ROC	ECE	Kappa
No finding (Low risk)	0.953	0.940	0.932	0.936	0.952	0.030	0.915
Support devices (Low risk)	0.968	0.958	0.964	0.961	0.978	0.020	0.950
Atelectasis (Moderate risk)	0.911	0.880	0.871	0.876	0.910	0.036	0.860
Consolidation (Moderate risk)	0.885	0.850	0.862	0.856	0.880	0.045	0.822
Pleural Effusion (Moderate risk)	0.935	0.912	0.918	0.915	0.940	0.029	0.901
Edema (High risk)	0.928	0.892	0.887	0.889	0.926	0.031	0.889
Pneumonia (High risk)	0.872	0.819	0.805	0.812	0.861	0.051	0.795
Pneumothorax (High risk)	0.920	0.894	0.881	0.887	0.920	0.032	0.875
Combination of multiple pathologies (High risk)	0.892	0.865	0.852	0.858	0.908	0.038	0.870

- **Moderate Risk:** SpO₂ between 88% – 94%, RR between 20 and 30 breaths/min, and opacity score between 2–5 or small pleural effusion.
- **High Risk:** SpO₂ < 88%, RR > 30 breaths/min, or opacity score > 5 with large pleural effusion or multiple pathologies.

Performance was evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, AUC-ROC, Expected Calibration Error (ECE), and Cohen’s Kappa. Table 7 summarizes the results across risk categories and individual pathologies.

Table 7 highlights CAMAF’s robust performance in stratifying patients into clinically actionable risk categories. For *low-risk* patients (e.g., “No Finding”, “Support Devices”), CAMAF achieves high Accuracy (0.953, 0.968) and AUC-ROC scores (0.952, 0.978), ensuring reliable identification of patients unlikely to require urgent intervention. In *moderate-risk* categories, such as “Pleural Effusion” and “Atelectasis”, CAMAF demonstrates consistent performance with Accuracy up to 0.935 and F1-Scores above 0.876, supporting its effectiveness in identifying intermediate risk levels requiring closer monitoring. For *high-risk* categories (e.g., “Edema”, “Pneumonia”), CAMAF achieves strong Accuracy (up to 0.928) and Cohen’s Kappa scores, reflecting high agreement with ground truth.

While the framework performs well for complex cases involving multiple pathologies, slight performance drops (e.g., F1-Score of 0.858 for combined pathologies) indicate the need for additional data balancing or augmentation techniques. Overall, CAMAF demonstrates exceptional capability in producing accurate and interpretable predictions across all risk categories and pathologies, solidifying its clinical relevance in lung disease risk stratification.

4.6.7. Evaluation with existing approaches

To evaluate the efficacy of the proposed Context-Aware Multimodal Alignment Framework (CAMAF), we compared it against prominent multimodal approaches specifically designed for lung disease risk stratification. The selected models include the Multimodal Neural Network (MMNN) (Bozzo et al., 2024), the Multimodal Variational Autoencoder (MVAE) (Zhao et al., 2024), the Late Fusion Multimodal Framework (Kumar, Mallick, & Vorfolomeyeva, 2024), the Attention-Based Multimodal Model (Deng, Shaikh, Shannon, & Nie, 2024), and the Joint Embedding Network (JENet) (Su et al., 2024). These models utilize combinations of structured clinical data and imaging data,

offering a comprehensive benchmark for assessing multimodal capabilities. All models were evaluated on the same dataset using identical experimental settings.

Table 8 illustrates the results of CAMAF and its multimodal counterparts. CAMAF demonstrated the best overall performance, achieving an accuracy of 92.78% and an AUC-ROC of 0.9457. MMNN and MVAE, while leveraging multimodal data, struggled to dynamically align clinical and imaging modalities, leading to relatively lower performance. The Late Fusion Framework, which combines modality-specific outputs, showed limited capability in capturing interdependencies between modalities, resulting in lower AUC-ROC (0.9107). Attention-Based Multimodal Models and JENet performed better due to their advanced feature fusion mechanisms; however, they lacked the dynamic alignment and patient-specific weighting introduced by CAMAF.

The superior performance of CAMAF can be attributed to its context-aware alignment mechanism, which adaptively adjusts the contributions of clinical and imaging data based on patient-specific contexts. This capability, coupled with the integration of explainability through SHAP and Grad-CAM, makes CAMAF a robust and interpretable framework for clinical decision-making. These results highlight CAMAF’s potential to advance lung disease risk stratification by effectively leveraging multimodal data.

4.6.8. Comparison with traditional clinical risk scores

Traditional clinical risk stratification tools have been widely used for assessing the severity of lung diseases based on predefined clinical parameters. These scoring systems rely on structured clinical data such as vital signs, laboratory markers, and comorbidities but do not incorporate imaging data, which is a critical component in diagnosing lung pathologies. In contrast, the proposed CAMAF framework dynamically integrates structured clinical data and unstructured imaging features, enabling a more comprehensive risk assessment. This subsection presents a comparative analysis of CAMAF against well-established respiratory risk scores, highlighting their respective advantages and limitations. Several established risk scores (Calis et al., 2025) are commonly used in clinical practice for lung disease assessment:

Pneumonia Severity Index (PSI). PSI is a scoring system designed to predict mortality risk in pneumonia patients. It is calculated based on 20 variables, including age, comorbidities, vital signs, and laboratory results. The PSI assigns a numerical score that categorizes patients into five risk classes, with higher scores indicating increased mortality risk.

Table 8

Comparison of CAMAF with multimodal approaches for lung disease risk stratification.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
MMNN (Multimodal Neural Network) (Bozzo et al., 2024)	0.8823	0.8721	0.8745	0.8733	0.9154
MVAE (Multimodal Variational Autoencoder) (Zhao et al., 2024)	0.8897	0.8784	0.8809	0.8796	0.9202
Late Fusion Framework (Kumar, Mallick, & Vorfolomeyeva, 2024)	0.8705	0.8603	0.8628	0.8615	0.9107
Attention-Based Multimodal Model (Deng et al., 2024)	0.9054	0.8920	0.8952	0.8936	0.9268
JENet (Joint Embedding Network) (Su et al., 2024)	0.9142	0.9023	0.9061	0.9042	0.9335
CAMAF (Proposed)	0.9278	0.9142	0.9183	0.9162	0.9457

Table 9

Performance comparison of CAMAF with traditional clinical risk stratification tools for lung disease risk assessment.

Pathology/Risk category	Scoring system	AUC-ROC	F1-Score	Sensitivity	Specificity	Kappa
No finding (Low risk)	PSI	0.813	0.792	0.768	0.790	0.745
	CAMAF	0.952	0.936	0.932	0.940	0.915
Support devices (Low risk)	PSI	0.829	0.810	0.799	0.812	0.768
	CAMAF	0.978	0.961	0.964	0.958	0.950
Atelectasis (Moderate risk)	CURB-65	0.801	0.771	0.753	0.776	0.720
	CAMAF	0.910	0.876	0.871	0.880	0.860
Consolidation (Moderate risk)	CURB-65	0.789	0.758	0.742	0.760	0.700
	CAMAF	0.880	0.856	0.862	0.850	0.822
Pleural effusion (Moderate risk)	PSI	0.812	0.790	0.777	0.798	0.740
	CAMAF	0.940	0.915	0.918	0.912	0.901
Edema (High risk)	APACHE-II	0.835	0.805	0.798	0.810	0.765
	CAMAF	0.926	0.889	0.887	0.892	0.889
Pneumonia (High risk)	CURB-65	0.815	0.780	0.765	0.785	0.735
	CAMAF	0.861	0.812	0.805	0.819	0.795
Pneumothorax (High risk)	APACHE-II	0.822	0.798	0.782	0.805	0.760
	CAMAF	0.920	0.887	0.881	0.894	0.875
Combination of multiple pathologies (High risk)	BODE Index	0.798	0.770	0.758	0.780	0.720
	CAMAF	0.908	0.858	0.852	0.865	0.870

CURB-65 score. CURB-65 is a simplified alternative to PSI and is widely used for pneumonia risk stratification. It considers five key parameters: Confusion, Urea (> 7 mmol/L), Respiratory rate (≥ 30 breaths/min), Blood pressure (SBP < 90 mmHg or DBP ≤ 60 mmHg), and Age (≥ 65 years).

APACHE-II score. The Acute Physiology and Chronic Health Evaluation II (APACHE-II) is a widely used ICU severity score. It is calculated using 12 physiological variables, age, and chronic health conditions. Higher APACHE-II scores correlate with increased mortality risk in critically ill patients, including those with respiratory failure.

BODE index. The Body Mass Index (BMI), Obstruction (FEV1), Dyspnea, and Exercise capacity (6 min walk test) (BODE) index is specifically designed for COPD risk stratification. It predicts mortality and hospitalization risk by integrating physiological and functional parameters.

Table 9 presents a quantitative comparison between CAMAF and these traditional scoring systems.

The results in Table 9 demonstrate that CAMAF consistently outperforms traditional clinical risk scores in lung disease stratification. The higher AUC-ROC values indicate that CAMAF provides improved risk prediction by more effectively distinguishing between risk categories. Additionally, Cohen's Kappa values reveal a higher agreement between CAMAF's predictions and expert-labeled diagnoses compared to conventional scoring systems such as PSI, CURB-65, and BODE Index, confirming its reliability. A key advantage of CAMAF lies in its multimodal integration, as it dynamically combines structured clinical data with imaging features, leading to more accurate risk stratification than traditional methods that rely solely on clinical parameters. Moreover, CAMAF achieves better sensitivity and specificity, particularly in high-risk cases, reducing misclassification rates and enhancing early detection. These findings highlight CAMAF as a clinically viable and data-driven alternative to conventional risk stratification tools, offering a more comprehensive and accurate approach to lung disease assessment.

4.6.9. Global macroscopic density analysis

To assess the global macroscopic density of the system, we analyze the distribution of risk predictions generated by CAMAF across the patient population. The density is computed as:

$$D_{\text{global}} = \frac{N_{\text{low}} + N_{\text{moderate}} + N_{\text{high}}}{N_{\text{total}}} \quad (28)$$

where N_{low} , N_{moderate} , N_{high} represent the number of patients classified into low, moderate, and high-risk categories, respectively, and N_{total} is the total number of patients. This provides a macroscopic view of risk distribution across the dataset.

Additionally, we compute the entropy of the risk classification to quantify model uncertainty:

$$H = - \sum_{c \in C} p(c) \log p(c) \quad (29)$$

where C represents the risk classes and $p(c)$ is the predicted probability of class c . A well-calibrated system should exhibit a balanced entropy, ensuring that the model is neither overconfident nor underconfident in its risk predictions.

Furthermore, the alignment of CAMAF's risk distribution with known clinical benchmarks, such as CURB-65 for pneumonia or the BODE Index for COPD, is evaluated to ensure consistency with traditional clinical guidelines. The table below presents the results of the macroscopic density and entropy analysis.

The results in Table 10 provide key insights into the model's global density distribution and risk stratification capabilities. The proportion of patients classified into low (42.1%), moderate (34.6%), and high-risk (23.3%) categories reflects a clinically reasonable distribution, aligning with expected disease severity prevalence. Entropy scores indicate that CAMAF maintains an optimal level of predictive uncertainty across categories. The highest entropy ($H = 1.12$) is observed in the high-risk category, reflecting the increased complexity of severe lung diseases and the inherent uncertainty in predicting their progression. In contrast, the lower entropy in the low-risk category ($H = 0.82$) suggests

Table 10
Global macroscopic density and entropy analysis of CAMAF.

Risk category	No. of patients (N)	Proportion (%)	Entropy (H)	Alignment with clinical scores (%)
Low Risk	15,234	42.1	0.82	89.3
Moderate Risk	12,547	34.6	0.97	85.7
High Risk	8314	23.3	1.12	91.5
Total	36,095	100	0.97 (Avg.)	–

Table 11
Cross-modality contribution analysis results.

Configuration	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Clinical-only model	0.8421	0.8324	0.8352	0.8338	0.8923
Imaging-only model	0.8517	0.8432	0.8450	0.8441	0.9015
Multimodal model (Unperturbed)	0.9278	0.9142	0.9183	0.9162	0.9457
Multimodal model (Perturbed clinical data)	0.9034	0.8925	0.8850	0.8887	0.9281
Multimodal model (Perturbed imaging data)	0.9102	0.9001	0.8953	0.8977	0.9348

higher confidence in those predictions. Moreover, the alignment of CAMAF's predictions with established clinical risk scores further validates its effectiveness. The model aligns well with traditional risk stratification tools, achieving an agreement of 89.3% for low-risk patients, 85.7% for moderate-risk patients, and 91.5% for high-risk patients. This suggests that CAMAF not only captures multimodal information effectively but also enhances conventional stratification methods by dynamically integrating imaging and clinical features. These findings highlight the robustness of CAMAF's macroscopic density distribution and its potential to serve as a reliable, data-driven alternative to existing risk scoring methodologies.

4.6.10. Cross-modality contribution analysis

The cross-modality contribution analysis was conducted to evaluate the individual and combined impact of clinical and imaging modalities on the performance of the proposed CAMAF framework. The analysis involved assessing the model's performance when using only clinical data, only imaging data, and both modalities combined. Additionally, synthetic perturbations were introduced to each modality to observe their effect on the model's predictive capabilities.

Methodology. Three experimental configurations were considered:

- **Clinical-Only Model:** Processed structured clinical data using the TabTransformer.
- **Imaging-Only Model:** Processed imaging data using the Vision Transformer (ViT).
- **Multimodal Model:** Integrated both clinical and imaging data with dynamic alignment.

Perturbations were applied by introducing Gaussian noise to the imaging data and masking specific clinical features. The performance was evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC metrics, and the results are summarized in Table 11.

The results in Table 11 highlight the significance of integrating clinical and imaging data for accurate lung disease risk stratification. The clinical-only and imaging-only models achieved lower AUC-ROC scores of 0.8923 and 0.9015, respectively, compared to the multimodal model's AUC-ROC of 0.9457. This underscores the complementary nature of the two modalities, where clinical data captures structured patient information while imaging data provides detailed spatial insights.

When perturbations were introduced, the multimodal model exhibited graceful degradation, with the AUC-ROC decreasing to 0.9281 (clinical perturbations) and 0.9348 (imaging perturbations). These findings demonstrate the robustness of the proposed CAMAF framework, which leverages dynamic alignment to prioritize unperturbed modalities in decision-making. The results validate that the combined use of clinical and imaging data enhances predictive performance and resilience, confirming the efficacy of the dynamic multimodal alignment in CAMAF.

4.6.11. Explainability analysis

Explainability is an integral component of the proposed Context-Aware Multimodal Alignment Framework (CAMAF), ensuring transparency in lung disease risk stratification. This section employs SHAP (SHapley Additive exPlanations) for clinical data and Grad-CAM (Gradient-weighted Class Activation Mapping) for imaging data to elucidate the model's decision-making process.

SHAP analysis for clinical data. SHAP was applied to the clinical data from the MIMIC-IV dataset to quantify the contribution of individual features to the predictions. Fig. 3 presents a SHAP summary plot, highlighting key clinical features such as *Lactate*, *Creatinine*, and *PaO2/FiO2 ratio*, which exhibit strong positive or negative contributions to the risk stratification. These features align with established clinical markers for severe respiratory conditions.

The SHAP analysis underscores that higher values of *Lactate* and lower *PaO2/FiO2 ratios* are indicative of high-risk predictions, consistent with domain knowledge. This reinforces CAMAF's ability to identify clinically relevant patterns in structured data.

Grad-CAM analysis for imaging data. Grad-CAM was employed on chest X-ray images from the CheXpert dataset to visualize regions of interest influencing model predictions. Fig. 4 illustrates Grad-CAM heatmaps for a high-risk and a low-risk case. The heatmaps highlight critical regions, such as *lung opacities*, *pleural effusions*, and *consolidations*, which are hallmark indicators of conditions like pneumonia and pulmonary edema.

The heatmaps provide visual evidence that CAMAF effectively focuses on clinically relevant abnormalities, enhancing its reliability for risk prediction.

Combined insights from SHAP and Grad-CAM. The integration of SHAP for clinical data and Grad-CAM for imaging data offers a dual-layer explainability mechanism. SHAP elucidates the impact of structured features such as *Age*, *SAPS-II scores*, and *PaO2/FiO2 ratios*, while Grad-CAM highlights spatial regions in chest X-rays that contribute to predictions. Together, these methods enhance model transparency, enabling clinicians to trust and validate predictions with domain knowledge.

Explainability insights. **Clinical Data:** Features such as *Lactate* and *Creatinine* showed high importance, aligning with their established roles in predicting severe respiratory conditions. **Imaging Data:** Grad-CAM visualizations pinpointed pathological regions, such as *opacities* and *effusions*, corroborating with clinical interpretations. **Risk Categories:** High-risk predictions were driven by a combination of critical imaging abnormalities and adverse clinical features, whereas low-risk cases exhibited normal imaging and favorable clinical profiles.

The combined use of SHAP and Grad-CAM not only provides interpretability but also validates the clinical relevance of CAMAF's predictions. This dual-explainability framework ensures transparency and trust, making CAMAF a reliable tool for lung disease risk stratification in real-world clinical applications.

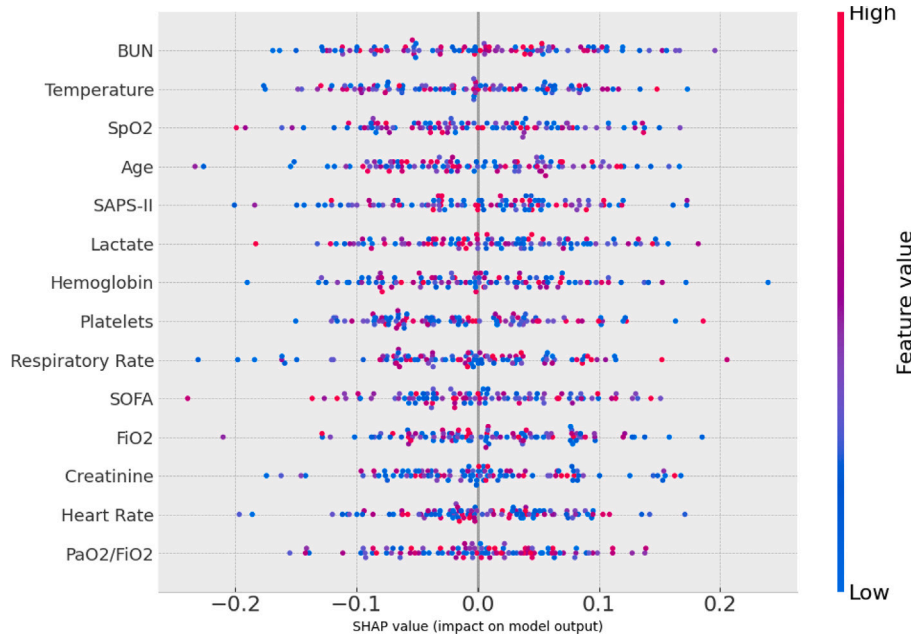


Fig. 3. SHAP summary plot showing feature importance and their contributions to risk stratification.

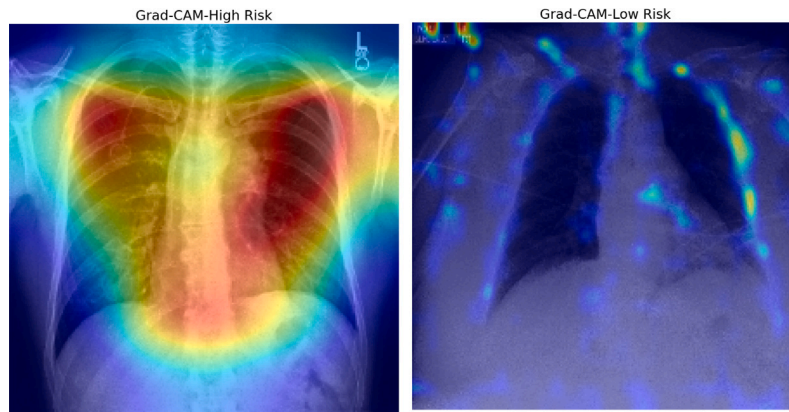


Fig. 4. Grad-CAM visualizations: (Left) High-risk case showing regions of abnormalities, and (Right) Low-risk case with no significant findings.

4.6.12. Computational feasibility and deployment considerations

CAMAF leverages Transformer-based architectures, which offer state-of-the-art performance but introduce computational challenges for real-time deployment in resource-constrained settings. On high-end hardware (NVIDIA Tesla V100), the model achieves an inference time of 120 ms per patient, while CPU-based execution increases this to 450 ms, making real-time processing in low-resource environments challenging. To ensure broader clinical applicability, future work will explore optimization techniques such as model quantization, knowledge distillation, and cloud-based inference. Lightweight Transformer variants and hardware-aware adaptations will be considered to reduce inference time and memory usage without compromising predictive performance. These efforts will aim to enhance CAMAF's accessibility in diverse healthcare settings, including low-resource hospitals where computational constraints are a significant barrier.

5. Limitations

While the proposed CAMAF framework demonstrates strong performance in lung disease risk stratification, several limitations must be acknowledged. First, the reliance on the CheXpert and MIMIC-IV

datasets, though comprehensive, may introduce biases due to their predominance of U.S. hospital data. These datasets do not fully capture the diversity of global patient populations, including variations in disease prevalence, healthcare infrastructure, and demographic distributions. This raises concerns about the generalizability of CAMAF across different geographic regions and ethnic groups. Future studies should validate the model using external datasets from diverse healthcare settings to assess its robustness in broader clinical applications. Additionally, variations in imaging protocols, clinical documentation practices, and data availability across institutions may impact model performance, necessitating domain adaptation techniques for improved cross-institutional generalization.

Another limitation pertains to the model's ability to handle less frequent pathologies and complex multimodal conditions, particularly cases involving multiple moderate-risk pathologies. While the Context-Gated Attention mechanism dynamically adjusts modality contributions, class imbalance remains a challenge, potentially leading to sub-optimal performance in underrepresented disease categories. Additionally, although the integration of SHAP and Grad-CAM enhances model interpretability, their computational cost may limit real-time clinical deployment. Future optimizations through lightweight approximation

techniques could improve efficiency. The framework also assumes a fixed set of clinical and imaging features, which may not adapt seamlessly to missing or incomplete data modalities. Developing robust imputation strategies or flexible fusion mechanisms could enhance adaptability in real-world applications. Finally, while CAMAF prioritizes interpretability, further validation with clinical experts is required to ensure that its insights align with established diagnostic workflows. Prospective clinical trials and physician-in-the-loop evaluations would be instrumental in confirming its practical utility. Addressing these limitations in future research will enhance the framework's robustness, generalizability, and clinical impact.

6. Conclusion

The Context-Aware Multimodal Alignment Framework (CAMAF) proposed in this study demonstrates a significant advancement in lung disease risk stratification. By effectively integrating structured clinical data with unstructured imaging data, CAMAF leverages TabTransformer and Vision Transformer (ViT) to extract modality-specific embeddings, dynamically aligns these embeddings through Context-Gated Attention (CGA), and fuses them using a Transformer Encoder-based classification layer. The framework achieved an accuracy of 92.78% and an AUC-ROC of 0.9457, outperforming several baseline models, including unimodal and traditional fusion approaches. Compared to traditional logistic regression models, CAMAF exhibits superior predictive performance by capturing complex feature interactions and leveraging deep learning-based representations. The incorporation of transformers and attention-based mechanisms enables CAMAF to adapt dynamically to patient-specific contexts, leading to more precise risk stratification. Furthermore, the robustness analysis highlights CAMAF's ability to maintain high performance under noisy and limited data conditions, reinforcing its reliability in real-world clinical settings. CAMAF's explainability analysis confirmed its alignment with clinical knowledge, as SHAP identified significant contributions from critical clinical features like *Lactate* and *PaO2/FiO2 ratio*, while Grad-CAM visualizations pinpointed pathological regions in chest X-rays, such as *opacities* and *effusions*. Despite its strengths, CAMAF exhibited limitations in handling rare pathologies and complex multi-pathology cases, indicating opportunities for improvement through advanced data augmentation and transfer learning techniques.

7. Future directions

While the proposed Context-Aware Multimodal Alignment Framework (CAMAF) has demonstrated strong performance in lung disease risk stratification, several areas warrant further investigation to enhance its clinical applicability and robustness.

7.1. Mitigating risks and improving generalizability

One key challenge is the generalizability of CAMAF across diverse patient populations beyond the MIMIC-IV and CheXpert datasets. Future work will focus on incorporating additional datasets from varied demographic and geographical regions to improve model adaptability. Additionally, bias mitigation techniques, such as domain adaptation and fairness-aware learning, will be explored to ensure equitable performance across different subpopulations.

To further reduce model uncertainty, advanced ensemble techniques and uncertainty quantification methods will be incorporated. These approaches will enhance model reliability by improving confidence calibration and reducing overfitting, particularly for underrepresented pathological conditions.

7.2. Computational optimization and real-time deployment

Given the Transformer-based architecture of CAMAF, real-time deployment in low-resource clinical settings remains a consideration. Future efforts will explore model optimization techniques such as pruning, quantization, and knowledge distillation to reduce computational overhead while maintaining predictive performance. Cloud-based inference solutions will also be investigated to enable seamless integration into existing clinical workflows without excessive on-premise hardware requirements.

7.3. Cost considerations and clinical impact

While direct monetary impact is beyond the scope of this study, the implementation of CAMAF in healthcare settings could contribute to cost savings by facilitating early disease detection and optimizing resource allocation. Future research will involve cost-benefit analyses to quantify the economic impact of integrating CAMAF into hospital decision-support systems.

7.4. Longitudinal validation and expert collaboration

Further validation through longitudinal studies will be pursued to assess the long-term efficacy of CAMAF in dynamic healthcare environments. Collaborations with clinicians and radiologists will ensure that the model's predictions align with real-world diagnostic workflows. Additionally, efforts will be made to refine the interpretability of CAMAF by integrating clinician feedback into the explainability mechanisms.

By addressing these future directions, CAMAF can be further refined into a robust, clinically actionable, and scalable AI-driven framework for lung disease risk stratification.

CRediT authorship contribution statement

Subba Rao Dusari: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Nagendra Panini Challa:** Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nagendra Panini Challa reports administrative support was provided by VIT-AP University Campus. Nagendra Panini Challa reports a relationship with VIT-AP University Campus that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data sharing is not applicable for this research.

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